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Modular Material Handling Robot Platform

Abstract

An apparatus including a first base plate, where the first base plate is configured to have at least one linear drive component and/or at least one power coupling component connected to a top side of the first base plate, where the first base plate is configured to be located inside a vacuum chamber; and a plurality of rails or transport guides on the top side of the first base plate. An end of the first base plate includes at least one alignment feature configured to align an end of the first base plate to an end of a second base plate. The first base plate is configured to provide, in combination with the second base plate, a structural platform inside the vacuum chamber for a robot drive to move in the vacuum chamber along the plurality of transport guides.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/432,192, filed Feb. 5, 2024, which is a divisional application of U.S. patent application Ser. No. 16/788,973, filed Feb. 12, 2020, and claims priority under 35 USC 119(e) to U.S. provisional application No. 62/805,532, filed Feb. 14, 2019, which are hereby incorporated by reference in their entireties.

BACKGROUND

Technical Field

[0002] The example and non-limiting embodiments relate generally to a substrate transport assembly.

Brief Description of Prior Developments

[0003] Robots for transporting substrates are known. Linear drive systems for transporting a substrate transport robot is also known such as described in U.S. patent publication numbers US 2016/0229296 A1, US 2013/0071218 A1, US 2015/0214086 A1, and US 2017/0028546 A1 which are hereby incorporated by reference in their entireties.

SUMMARY

[0004] The following summary is merely intended to be exemplary. The summary is not intended to limit the scope of the claims.

[0005] In accordance with one aspect, an example embodiment is provided in an apparatus comprising: a first base plate, where the first base plate is configured to have at least one linear drive component and/or at least one power coupling component connected to a top side of the first base plate, where the first base plate is configured to be located inside a vacuum chamber; and a plurality of rails on the top side of the first base plate, where an end of the first base plate comprises at least one alignment feature configured to align an end of the first base plate to an end of a second base plate, where the first base plate is configured to provide, in combination with the second base plate, a structural platform inside the vacuum chamber for a robot drive to move in the vacuum chamber along the plurality of rails.

[0006] In accordance with another aspect, an example method is provided comprising: connecting a first base plate assembly inside a vacuum chamber, where the first base plate assembly comprises a first base plate and a first plurality of rails on a top side of the first base plate; and connecting a second base plate assembly inside the vacuum chamber, where the second base plate assembly comprises a second base plate, where an end of the second base plate is connected to an end of the first base plate, where the first base plate assembly comprises at least one alignment feature configured to align the end of the first base plate to the end of the second base plate, where the first base plate is configured to provide, in combination with the second base plate, a structural platform inside the vacuum chamber for a robot drive to move in the vacuum chamber along the first plurality of rails.

[0007] In accordance with another aspect, a method is provided comprising: providing a base plate; providing a plurality of rails on a top side of the base plate; providing an end of the base plate with at least one alignment feature configured to align the end of the first base plate to an end of a second base plate; connecting to the top side of the base plate at least one of: a power coupling

component configured to transfer power to another power coupling component on a robot drive, where the transfer of power is with induction, or a linear motor component configured to provide a magnetic field to move the robot drive along the plurality of rails, where the base plate is configured to be connected inside a vacuum chamber with the second base plate to provide a structural platform inside the vacuum chamber for the robot drive to move in the vacuum chamber along the first plurality of rails.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The foregoing aspects and other features are explained in the following description, taken in connection with the accompanying drawings, wherein:

[0009] FIG. 1 is a schematic top view of a substrate processing apparatus comprising features as described herein;

[0010] FIG. 2 is a perspective view of a substrate transport apparatus of the substrate processing apparatus shown in FIG. 1;

[0011] FIG. 3 is a perspective view of the substrate transport apparatus shown in FIG. 2;

[0012] FIG. 4 is an end view of the perspective view of a substrate transport apparatus shown in FIGS. 2-3 shown inside a transport chamber of the substrate processing apparatus;

[0013] FIG. 5 is a side view of a robot of the substrate transport apparatus shown in FIGS. 1-4;

[0014] FIG. 6 is a top view of the robot shown in FIG. 5;

[0015] FIG. 7 is a schematic sectional view of the robot shown in FIGS. 5-6;

[0016] FIG. 8 is an end view of the substrate transport apparatus inside the transport chamber of the substrate processing apparatus;

[0017] FIG. 9 is a diagram illustrating various systems used with the apparatus shown in FIG. 1;

[0018] FIG. 10 is a schematic top view of the substrate transport apparatus inside the transport chamber of the substrate processing apparatus shown in FIG. 1;

[0019] FIG. 10A is an end sectional view of the substrate transport apparatus inside the transport chamber of the substrate processing apparatus shown in FIG. 10;

[0020] FIG. 11 is an end view of the substrate transport apparatus inside the transport chamber as shown in FIG. 10;

[0021] FIG. 12 is a perspective view showing one of the magnetic drivers and the driven member of the linear motor shown in FIGS. 10-11;

[0022] FIG. 13 is a perspective view of the magnetic driver shown in FIG. 12;

[0023] FIG. 14 is a perspective view of a power module used with the apparatus shown in FIGS. 10-11;

[0024] FIG. 14A is a perspective view illustrating the relative layout of a plurality of the power modules shown in FIG. 14;

[0025] FIG. 15 is a diagram illustrating a power coupling using the power modules shown in FIGS. 14 and 14A;

[0026] FIG. 16 is an isometric representation of one exemplary embodiment coupling shown in FIG. 15;

[0027] FIG. 17 is a schematic view illustrating some components of an optical communication system shown in FIGS. 10-11;

[0028] FIG. 18 is a perspective view of the substrate transport apparatus on the bottom wall of the transport chamber;

[0029] FIG. 18A is a perspective view of an alternate embodiment of the substrate transport apparatus on the bottom wall of the transport chamber;

[0030] FIG. 19 is a schematic top view of a substrate processing apparatus comprising features as

described herein;

[0031] FIG. **20A** is a schematic side view of a material-handling robot platform comprising features as described herein;

[0032] FIG. **20B** is an end view of the platform shown in FIG. **20A**;

[0033] FIG. **20C** is a top plan view of the platform shown in FIG. **20A-20B**;

[0034] FIG. **20D** is a partial perspective view of an end of one of the base plate assemblies of the platform shown in FIGS. **20A-20C**;

[0035] FIG. **20E** is a partial perspective view of a rail or maglev guide integrally formed on a base plate assembly;

[0036] FIG. **20F** is a partial cross sectional view of one of the base plate assemblies showing some of the apertures and seals;

[0037] FIG. **21A** is a schematic side view of a material-handling robot platform and cross section of a chamber comprising features as described herein;

[0038] FIG. **21B** is an end cross sectional view of the platform shown in FIG. **21A**;

[0039] FIG. **21C** is a top cross sectional view of the platform shown in FIG. **21A-21B**;

[0040] FIG. **22A** is a schematic side view of a material-handling robot platform and cross section of a chamber comprising features as described herein;

[0041] FIG. **22B** is an end cross sectional view of the platform shown in FIG. **22A**;

[0042] FIG. **22C** is a top cross sectional view of the platform shown in FIG. **22A-22B**;

[0043] FIG. **23A** is a schematic side view of a material-handling robot platform and cross section of a chamber comprising features as described herein;

[0044] FIG. **23B** is an end cross sectional view of the platform shown in FIG. **23A**;

[0045] FIG. **23C** is a top cross sectional view of the platform shown in FIG. **23A-23B**;

[0046] FIG. **24** is a diagram illustrating use of multiple sensors to account for transitions between linear maglev tracks/guides and maglev gap sensor targets;

[0047] FIG. **25** (FIGS. **25A-25C** collectively), FIG. **26** (FIGS. **26A-26C** collectively), FIG. **27** (FIGS. **27A-27C** collectively), FIG. **28** (FIGS. **28A-28C** collectively), FIG. **29** (FIGS. **29A-29C** collectively), FIG. **30** (FIGS. **30A-30C** collectively) are diagrams illustrating various different example embodiments illustrating examples of linear robot control systems.

DETAILED DESCRIPTION OF EMBODIMENTS

[0048] Referring to FIG. **1**, there is shown a schematic top plan view of an apparatus **10** incorporating features of an example embodiment. Although the features will be described with reference to the example embodiments shown in the drawings, it should be understood that features can be embodied in many alternate forms of embodiments. In addition, any suitable size, shape or type of elements or materials could be used.

[0049] The apparatus **10**, in this example, is a substrate processing apparatus. The substrate processing apparatus **10** generally comprises a substrate transport apparatus **12** (also referred to as a linear vacuum robot), multiple substrate processing chambers **14**, a transport chamber **15**, an equipment front end module (EFEM) **16**, and substrate cassette elevators **18**. The transport chamber **15** may be maintained as a vacuum chamber or inert gas chamber for example. The transport apparatus **12** is located in the chamber **15** and is adapted to transport substrates **20**, such as semiconductor wafers or flat panel displays for example, between and/or among the chambers **14**, **15**, and the stationing transfer chambers or load locks **22**. The EFEM **16** is configured to transport the substrates **20** between the substrate cassette elevators **18** and stationing transfer chambers **22**. In this example, the EFEM has a robot **24** having a scara arm. The robot **24** is configured to linearly move in the EFEM as indicated by arrow A. However, any suitable type of EFEM could be provided. The apparatus **10** includes a controller **50**. The controller **50** comprises at least one processor **52** and at least one memory **54** comprising computer program code **56**. The controller **50** is configured to control operations of the various devices and robots of the apparatus **10**.

[0050] Referring also to FIGS. 2-4, perspective views and an end view of the substrate transport apparatus **12** are shown. In this example the substrate transport apparatus **12** comprises a robot **26**, a robot enclosure **28**, and a linear drive system **30**. FIGS. 2-3 show the substrate transport apparatus **12** on the bottom wall of the transport chamber **15** with the side and top walls of the transport chamber **15** not shown. FIG. 4 shows the substrate transport apparatus **12** on the bottom wall or floor of the transport chamber **15** with merely the top wall of the transport chamber **15** not shown. The robot **26** is connected to the robot enclosure **28**, and the robot enclosure **28** is movably relative to the chamber **15** by means of the linear drive system **30**. The linear drive system **30** comprises guides **32** on the floor of the transport chamber **15** and trucks **33** on the exterior bottom side of the enclosure. For a maglev type of linear drive system, the trucks may be separated from the guides **32** by a spacing maintained by magnetism/electro-magnetism. For a non-maglev type of linear drive, the guides may comprise rails where the trucks ride on the rails. In an alternate example the guides may be maglev transport guides or tracks. In one example, the trucks **33** may have wheels attached to the bottom side of the enclosure which ride on the rails **32**. The linear drive system **30** is configured to move the robot enclosure **28** and, thus, move the substrate transport apparatus **12**, inside the transport chamber **15** along the rails **32** in a linear path as indicated by arrow B in FIG. 1. In alternate embodiments rather than an enclosure, the robot **26** may be attached to a non-enclosing slide or platform which has the trucks and rides on the rails/guides **32**. In another alternate example, the guides and trucks could be provided at lateral side walls of the chamber **15** in addition to, or as an alternative, to the floor of the chamber **15**.

[0051] Referring also to FIGS. 5-7, the robot **26** generally comprises a robot drive **34** and a robot arm assembly **36** connected to the robot drive **34**. The robot **26** shown in FIGS. 5-7 is merely one example of a robot having and effectors for moving substrates, and should not be considered as limiting. Any suitable type of robot, robot drive and robot arm could be provided. In this example the robot arm assembly **36** is a dual arm arrangement. Each of the arms **36a**, **36b** of the robot arm assembly **36** has arm links, pulleys, bands, and substrate support end effectors **38** which are driven by the coaxial drive shafts **40a**, **40b**, **40c** of the robot drive **34**. In an alternate example, the drive might include drive shafts which are not coaxially located relative to each other. The robot drive **34** has motors **42a**, **42b**, **42c** for each drive shaft, a vertical drive system **44** including a motor **46**, and various position encoders/sensors **48a**, **48b**, **48c** for the motors **42** and/or drive shafts **40**, and the vertical drive system **44**.

[0052] The robot **26** is mounted to the robot enclosure **28** such that almost all of the robot **26** is located inside the robot enclosure **28** except for the robot arm assembly **36**. This can be seen best in FIGS. 3-4 for example. In particular, the robot arm assembly **36** is located outside of the enclosure **28**, proximate the top side of the robot enclosure **28**, and the drive shafts **40a**, **40b**, **40c** extend through an aperture **58** (see FIG. 3) in the top side of the robot enclosure **28** from the rest of the drive **34** located inside the enclosure **28**. A seal **60** (see FIG. 3) is provided at the aperture **58** to seal the aperture **58**, but still allow the drive shafts **40a**, **40b**, **40c** to axially rotate and vertically move when moved by vertical drive system **44**. With this type of embodiment, the area inside the enclosure **28** may have a different environment than the area inside the chamber **15**. For example, the area inside the enclosure may merely be air at an atmospheric pressure, and the area inside the chamber **15** (outside of the enclosure **28**) may be a vacuum environment or an inert gas environment.

Nested Environments

[0053] FIG. 8 illustrates schematically a “nested” type of arrangement which may be used in one example. The robot enclosure **28** may provide a hermetically sealed robot enclosure for the robot drive **34**, except for the small portion of the drive shafts which extend through the aperture **58**. More specifically, all of the motors **42**, **44**, all of the sensors **48**, all of the electrical wiring, and most of the drive shafts **40** (see FIG. 7) are hermetically enclosed inside the robot enclosure **28**. All of the active electrical hardware of the robot drive **34**, including the sensors and motors, may be

maintained inside the hermetically sealed robot enclosure **28** at area **29** shown in FIG. **8**; separate from the environment outside the robot enclosure **28** in the area **17** of the transport chamber **15**. The area **17** may be maintained as a separate environment inside the transport chamber **15** separate from the outside atmospheric environment **13**. Thus, environment **29** is nested inside the environment **17**, where the environment **17** separates the environment **29** from the outside atmospheric environment **13** (normal atmosphere **13** outside of the chamber **15**). The environment **29** might be a vacuum environment, but does not need to be a vacuum environment even if the environment **17** is a vacuum environment. Because the environment **29** inside the robot enclosure **28** does not need to be a vacuum environment, this may help to prevent outgassing of vapors from components inside the robot enclosure **28**. This nesting of environments (area **29** nested inside area **17**) is particularly adapted to the robot enclosure **28** being movable inside the transport chamber **15** without breaching or being interrupted by the relative movement linearly along the length of the transport chamber **15**.

[0054] Referring also to FIG. **9**, in this example the apparatus **10** has the linear actuation system **30**, a power coupling system **62**, a data communications coupling system **64** and a heat transfer coupling system **66**. The power coupling system **62** may be used to provide electrical power to components inside the enclosure **28**. The data communications coupling system **64** may be used to send data signals to components inside the enclosure **28** (and/or through the enclosure to components in the arm) and/or to send data signals from the components inside the enclosure **28** to a data processor outside the chamber **15** (and/or through the enclosure from components in the arm). The heat transfer coupling system **66** may be used to transfer heat from components inside the enclosure **28** to outside the chamber **15**. In an alternate example embodiment, the power coupling system **62** and the data communications coupling system **64** may be at least partially combined to reduce the number of components inside the area **17** of the chamber **15**. In some embodiments, an apparatus may be provided which does not comprise all of the systems noted above.

[0055] In the example shown, all of these systems **30**, **62**, **64**, **66** are configured to not breach or interfere with the nested environments shown in FIG. **8**. In other words, the systems **30**, **62**, **64**, **66** are suitably constructed such that the environment inside the enclosure **28** remains separated from the environment inside the chamber **15** while all four systems **30**, **62**, **64**, **66** operate.

Linear Motor

[0056] Referring also to FIGS. **10** and **11**, the linear drive system **30** generally comprises a linear motor **70**. A linear motor **70** is an electric motor that has had its stator and rotor essentially “unrolled” so that instead of producing a torque (rotation) it produces a linear force along its length. An example of this is disclosed in US patent publication No. US 2015/0214086 A1 which is hereby incorporated by reference in its entirety. The figures show the linear motor located under the enclosure **28**. However, in alternate embodiments, one or more components of the linear drive might not be located under the enclosure **28**. For example, components of the linear motor and/or the linear drive system **30** might be located at lateral sides of the enclosure **28** and chamber **15**.

[0057] In one example embodiment the linear drive system **30** may comprise a maglev system utilizing a non-contact magnetically supported guidance subsystem. Referring also to FIGS. **12-13** the linear motor **70**, in the example shown, utilizes a modular design comprising stationary magnetic drivers **72** and a driven member **76**. The drivers **72** have a general “C” shape in this example. The general “C” shape forms a slot **73** with opposing electromagnets **74** on top and bottom sides of the slot **73**. The drivers **72** are stationarily mounted to the bottom wall of chamber **15** in a row, as seen best in FIG. **10**, and the driven member **76** is attached to the exterior of the bottom side of the robot enclosure **28**. In an alternate example the drivers **72** might be attached to the robot enclosure **28** and the rail **76** might be stationarily attached to the transport chamber **15**.

[0058] Shielding may be provided around linear rails. Referring also to FIG. **10A**, the transfer chamber **15** contains the robot with linear rails **32** and trucks **30**, and the linear rails and trucks may

be partially or completely enclosed in shields **31**. The shield purpose is to reduce or eliminate migration of undesirable particulate or contaminant material from rails or trucks upward into the transfer chamber, while allowing robot motion in the direction of the rail. Further, the shield, which may be a single part or an assembly of parts, prevents migration of material, particulates, broken wafer fragments from the transfer chamber into the operation area of the linear rails. This helps prevent contamination/clogging of the linear rail mechanism.

[0059] The transfer chamber may also include vent ports **16** and pump ports **17**, such that the vent ports are located substantially towards the top of the chamber **15** and the pump ports in the lower section or within the shielded area of shields **31**. Such an arrangement further prevents migration of contaminants from within the shielded area into the transfer chamber. When the vent port is active, the molecules from the vent will travel towards the pump port, carrying with them any trace of airborne contaminants. Even when the vent port is closed, active pumping will still pull any contaminants away from the chamber and into the shielded region.

[0060] As shown best in FIG. **12**, the driven member **76** extends into the slot of the “C” shaped drivers **72**. Thus, the drivers **72** may utilize the same rail **76**, extending from the bottom side of the robot enclosure **28**, in an autonomous fashion. In alternate aspects, other guide rail arrangements, more or less may be provided and mounted in alternate arrangements, for example, on side walls of the chamber **15**. As described above, the modular linear drive system may utilize a non-contact magnetically driven forcer subsystem. The subsystem may comprise one or more linear motor modules and one or more position feedback modules. With the use of the example described above, the enclosure **28** may be linearly moved inside the chamber **15**, as indicated by arrow B in FIG. **1**, without the need to pass any wires through the enclosure **28**. The linear motor **70** is entirely outside the enclosure **28** and, thus, provision of the linear motor **70** does not increase a risk of a breach between the two areas **17**, **29**. Also, because driven member **76** does not contact any of the drivers **72**, and driving of the driven member is merely magnetically controlled, there is less risk of contaminates from the driven member **76** and drivers **72** in the area **17**. Supply of electrical power to the drivers **72** may be controlled by the controller **50** shown in FIG. **1**. The drivers **72**, or portions of the drivers **72**, may be energized to create a magnetic field to move (such as accelerate and decelerate for example) the driven member **76** in linear directions B, and also magnetically positionally lock the enclosure **28** at fixed locations in front chambers **14** and **22** as needed.

[0061] Each linear motor module may have a stationary passive magnetic stainless steel part. The stationary passive magnetic stainless steel part may have a toothed portion that interacts with a corresponding primary forcer. The passive part may or may not also have magnets. Multiple supports may utilize the same secondary in an autonomous fashion. Each linear motor module may have a primary forcer coupled to supports where the primary forcer may have three phase windings and permanent magnets. In alternate aspects, permanent magnets may be provided as part of driven member for the purpose of offsetting gravity and dynamic loads. In alternate aspects, permanent magnets may be provided as part of one or more of the magnetic bearings for the purpose of offsetting gravity and dynamic loads. An example of a potential primary forcer and secondary topology is provided with the Siemens 1FN6 Design. In alternate aspects, any suitable forcer may be provided. The permanent magnets of the forcers may be provided as a component that both facilitates efficient generation of thrust (coupled with windings) and also offsets the payload such that the magnetic bearings minimize the use of power during normal operation. Here, the attractive force between the forcer and the corresponding passive rail may be set at a nominal gap such that the force offsets gravity-induced forces resulting in minimum power consumption. Further, the set point for the gap may be varied such that as the payload changes, the gap is adjusted such that the force offsets gravity induced forces resulting in minimum power consumption as the payload changes. For example, the gap on a left forcer may be varied independently of that of a right forcer. Voltage may be selectively applied by the advanced control subsystem to the magnetic coils of the primary forcer to produce thrust to the support relative to the stationary passive magnetic stainless

steel secondary. Each stationary passive magnetic stainless steel secondary may be mounted with teeth oriented vertically down such that the attractive force of the primary forcer's permanent magnets may offset the weight of the support and the payload to minimize the DC component that needs to be applied by the vertical coils of the non-contact magnetically supported guidance subsystem.

Power Coupling System

[0062] Referring also to FIG. 14, the power coupling system 62, in this example embodiment, is configured to use magnetic inductional resonant effects to transfer electrical power to the substrate transport apparatus 12, such as for use with components inside the enclosure 28. The power coupling system 62 generally comprises a power coupling or module 78 on the interior bottom wall of the transport chamber 15 and a power coupling or module 80 on the exterior bottom side of the robot enclosure 28. As seen best in FIGS. 14A, 4, 10 and 11, a plurality of the primary modules 78 may be arranged aligned in a row on the bottom wall of the transport chamber 15, and the secondary module 80 may be located above the row of primary modules 78 attached to the exterior bottom side of the enclosure 28. An example of induction power transfer for a substrate transport apparatus is disclosed in US patent publication No. US 2016/0229296 A1 which is hereby incorporated by reference in its entirety. The power coupling may be combined with a communication device.

[0063] Referring to FIG. 15, there is shown a block diagram representation of one example embodiment of the power coupling system 62. As depicted, the power coupling system 62 may include an AC power source 82, the power coupling with at least one primary module 78 and at least one secondary module 80, and an optional rectifier circuit 84. AC power 86 is supplied to the primary 78 where the primary 78 in turn provides power to the secondary 80 over gap 88 where the secondary 80 provides AC power 90 to the rectifier circuit 84. The secondary 80 moves with the robot enclosure 28 while the primary 78 remains stationary. However, in an alternate example, the primary and secondary modules may be configured so that they can move with respect to each other, including translation, rotation or a combination of the two. The primary and secondary may be primary or secondary cores or primary or secondary rails. In alternate aspects, any suitable combination or geometry may be provided. An electrical line(s) from the module 80 may extend through a sealed aperture in the enclosure 28 to supply electricity, such as to motors 42 of the robot and/or to a controller (such as a servo motor controller for example) and/or communication equipment and/or sensors inside the enclosure 28 or subsequently to devices in or on the robot arm assembly 36 for example. In an alternate embodiment, more than one module 80 might be provided, and might be on one or more lateral sides of the enclosure 28. Because modules 78, 80 do not contact one another, there is less risk of contamination inside the chamber 15 as the module 80 moves relative to the modules 78.

[0064] Referring now to FIG. 16, an isometric representation of one example alternate embodiment of the power coupling 62a is shown. As seen best in FIG. 14, the primary module(s) 78 may consist of a primary core or rail 94 and a primary winding or coil 96, which may be configured so that AC current, e.g., supplied by the AC power source 82, through the primary winding 96 produces alternating magnetic flux in the primary core or rail 94. The primary core or rail 94 may feature extensions 98 along the direction of relative motion B between the primary module 78 and secondary module 80. The secondary module 80 may consist of a secondary core or rail 100 and a secondary winding or coil 102 configured so that alternating magnetic flux in the secondary core or rail 100 induces voltage in the secondary winding 102. The secondary core or rail 100 may be arranged so that it may move as part of the secondary module along the extensions 98 of the primary core or rail 94 and so that magnetic flux may pass between the extensions of the primary core or rail and the secondary core or rail at shoe portions across an air gap 104 between the extensions 98 of the primary core or rail and the secondary core or rail 100. The output of the secondary winding 102 may be used as an AC power source directly, or, if DC power is required,

the secondary winding may feed the rectifier circuit **84** (such as inside the enclosure **28** for example), which in turn may serve as a DC power source **92**, as illustrated in FIG. **15**. The primary winding **96** and secondary winding **102** may feature substantially the same number of turns in order for the amplitude of the output voltage of the secondary winding to be substantially equal to the amplitude of the voltage supplied by the AC power source **82**. The number of turns of the secondary winding **102** may be higher than the number of turns of the primary winding **96** if higher output voltage is required. Conversely, the number of turns of the secondary winding **102** may be lower than that of the primary winding **96** if lower output voltage is required. The primary core **94** and secondary core **100** may be C-shaped, as depicted diagrammatically in FIG. **16**, E-shaped, or they can feature any suitable shape that allows for inductive coupling between the primary module **78** and the secondary module **80**. The extensions **98** of the primary core **94** may be straight to support straight-line motion between the primary module **78** and secondary module **80**, or they may be curved to support curved or rotary motion. The primary core **94**, extensions **98** and secondary core **100** may be produced from a soft magnetic material, for instance, silicon steel, soft magnetic composite, another material suitable to channel magnetic flux or a combination of such materials. Laminated construction may be utilized. Here, the primary and secondary may be considered as induction sections with the windings being coils. All the electrical wiring of the secondary module **80** may be located inside the robot enclosure **28** so that no wiring is located in the area **17**. All the electrical wiring of the primary modules **78** may be located outside the transport chamber **15** so that no wiring is located in the area **17**.

Data Communications Coupling System

[0065] The data communications coupling system **64**, in this example, comprises an optical communication system **106** which, as shown best in FIGS. **10-11** and **17**, includes a first member **108** and a second member **110**. The first member **108** is connected to the exterior bottom side of the robot enclosure **28**. In an alternate example the first member **108** may be inside the enclosure with an optical window through the enclosure, or may be located on a side of the enclosure other than the bottom side. The second member **110** is connected to a side wall of the transport chamber **15**. The second member **110** may be located outside of the area **17** and the transport chamber **15** may have a window for the two members **108**, **110** to nonetheless optically communicate with each other. In an alternate example, the second member **110** may be inside the chamber **15**.

[0066] The two members **108**, **110** may use one or more laser beams or other optical signals **112** to transmit and receive data signal in the varying distances between the enclosure **28** and the chamber **15** as the enclosure moves inside the chamber **15**. The data signals may be subsequently transmitted to and from the components inside the robot enclosure **28**, such as to control the robot **26** and/or linear drive system **30**, and for data from sensors of the substrate transport apparatus **12** to be sent to the controller **50**. All the wiring (electrical and/or optical) from the first member **108** may be located inside the robot enclosure **28** so that no wiring is located in the area **17**. All the wiring (electrical and/or optical) from the second member **110** may be located outside the transport chamber **15** so that no wiring is located in the area **17**. This reduces the risk of contamination inside the area **17**, such as from outgassing from those wires. The communication may be combined with power delivery.

Heat Transfer Coupling System

[0067] As noted above, the apparatus **10** may also comprise a heat transfer coupling system **66**. The heat transfer coupling system **66** may be used to provide a thermal management in regard to components of the substrate transport apparatus **12** inside the robot enclosure **28**; to transfer heat from inside the robot enclosure **28** to outside the transport chamber **15**. This may be particularly important if the area **17** is a vacuum environment with poor heat transfer capability. The moving robot enclosure **28** serves to house all of the advanced control subsystem which move with the substrate transport apparatus **12**. The moving robot enclosure **28** further serves to support the robotic transfer arm that cooperates with the moving support to transport one or more substrates

between locations. As there are active components coupled to the moving robot enclosure **28**, such as the motors **42** for example, the heat generated by the active components may be dissipated by a thermal management subsystem. For a moving support in vacuum, heat may be dissipated either by radiation or by transfer through a medium, for example through a gas or by coupling a bellows to the moving support and circulating gas or liquid coolant through a chiller. In the event of cooling by radiation alone (or combination radiation and convection), an allowable temperature difference between all or part of the moving portion and the chamber may be specified, for example, 50 degrees Celsius or otherwise. With the example embodiment shown in the figures, non-contacting interleaving fin-like structures **120**, **122** (see FIGS. 2-4 and **10-11**) may be employed to maximize opposing surface areas. High emissivity coatings may also be utilized to maximize surface area related heat transfer. An example of suitable coatings may be aluminum oxide, aluminum nitride or any suitable high emissivity coating. In alternate aspects, any suitable surface or coating may be provided. For a moving support in a gas or inert environment, heat may be dissipated either by radiation or convection or both. As there are active components coupled to the moving support **28**, power and communication may be transferred to the moving support subsystem with the power coupling **62** and the communication coupling **64** as described with the above example. Power and communication may be transferred to the moving support subsystem wirelessly as described with the above example, by inductive coupling, via service loop or a combination of these approaches. [0068] Active components coupled to the support may be potted with vacuum compatible potting or epoxy or alternately be hermetically sealed within the enclosure **28** or a combination of both. Examples of suitable moving support thermally sunk subsystems are disclosed in Hosek M., Hofmeister C., Low Variability Robot, U.S. patent application Ser. No. 13/618,117, Sep. 14, 2012, U.S. Patent Publication Number US 2013/0071218 A1, which is hereby incorporated by reference herein in its entirety. However, because of the use of the robot enclosure **28**, use of potting or epoxy may be reduced or eliminated.

Example Alternatives

[0069] Referring also to FIG. **18** an alternate example embodiment is shown. This embodiment is similar to that shown in FIGS. 2-4, but has a reduced form factor. The linear motor is enclosed, there are shield linear bearings, and a protected optical link. The heat transfer fins might not be needed under some scenarios. Referring also to FIG. **18A** another alternate example embodiment is shown. This embodiment is similar to that shown in FIG. **18**, but in this example the enclosure **28** does not comprise heat transfer fins. In this type of embodiment, the wall cooling may be provided in the wall of the enclosure **28** and/or the walls of the vacuum chamber, such as running coolant through the chamber walls when there are no fins for cooling. However, such a cooling system could be used in conjunction with fins. FIG. **18A** shows an example of cooling conduits **15a** in the wall of the chamber **15'**.

[0070] In some embodiments the cooling of the robot may be enhanced by cooling of the vacuum chamber walls. Heat can be transferred from the robot drive to the vacuum chamber wall through radiation or convection, or some combination of the two. Regardless of the mechanism of heat transfer from the robot drive to the vacuum chamber, the quantity and rate at which heat is transferred may be enhanced by cooling of the chamber walls. Cooling of the chamber walls may be achieved by several different methods, or a combination thereof. In one example embodiment the cooling may be achieved by integration of cooling channels or vents, directly or indirectly on the vacuum chamber walls. A continuous, or intermittent flow of a working fluid, which can be a liquid, or a gas, or a combination of two, such that the working fluid has an initial inlet temperature lower than the vacuum chamber wall temperature, may be employed to achieve cooling of the vacuum wall temperature. This results in enhanced cooling of the robot drive. In an alternative embodiment, thermoelectric cooling tiles may be employed to cool the surfaces of the vacuum chamber, such that heat is rejected at a higher temperature on the hot side of the thermoelectric tiles, thus enabling more effective cooling.

[0071] Referring also to FIG. 19, an alternate example embodiment is shown. In this example the system is used between two cluster tools **150a**, **150b**. The chamber **15** has load locks **152** at opposite ends which are coupled to the tools **150a**, **150b**. The robot enclosure **28** and attached robot **26** can move as indicated by arrow B for transporting substrates between the tool robots **154a**, **154b**. The robot may also be used in a vacuum EFEM as in FIG. 1.

Features

[0072] One of the features as described herein is the ability of the robot enclosure **28** to function as a cart to linearly move inside the transport chamber **15** carrying all necessary components for the substrate transport apparatus **12** and the only physical contact with the transport chamber **15** being on the rails **32**. However, even that contact with the rails **32** may be eliminated if a maglev system is used.

[0073] Another feature as described herein is that virtually all components of the substrate transport apparatus **12**, except for the robot arm assembly **36** and top portions of the robot drive shafts, may be isolated inside the robot enclosure **28** within the area **29**. The area **29** may have an atmosphere which is not a vacuum; even if the area **17** is in vacuum. This allows the robot **26** to not be a vacuum compatible robot (a non-vacuum compatible robot). With a non-vacuum compatible robot outgassing is not a significant factor in its design and manufacture. Thus, without having to provide a low or nil outgassing design, a non-vacuum compatible robot is less expensive than a vacuum compatible robot. The robot enclosure **28** may also have a window for the optical communications such that the member **108** may be located totally inside of the robot enclosure **28**. The power coupling **80** may also be located totally inside of the robot enclosure **28** with a portion of the robot enclosure **28** having a magnetically transparent section for the two power couplings **78**, **80** to function properly with induction. The transport chamber **15** may be provided with a linear encoder **156** as shown in FIG. 10 to sense the linear location of the robot enclosure **28** on the rails **32**. All communications with the components in and/or on the robot enclosure **28** may be done optically or wirelessly, or through a power coupling, such that no communication electrical wires or optical wires need to traverse the area **17**.

[0074] With features as described herein, a robot drive **34** may be sealed by a hermetically sealed robot enclosure **28**. Thus, the transport chamber **15** might only need the rails **32** for the substrate transport apparatus **12** to move along, and no other direct physical contacts being made by the transport chamber **15** to the robot enclosure **28** or the substrate transport apparatus **12**. Even with a non-maglev embodiment, a gap or spacing may always be provided between the transport chamber **15** and the substrate transport apparatus **12** at all other locations except at the rails **32**. This reduces contamination of the area **17** by the substrate transport apparatus **12** because of such non-contact.

Modular Platform

[0075] Features as described herein may be used to provide a material-handling platform with a traversing vacuum-environment robot which is modular. A modular system may allow for scalability and eliminate the need to manufacture and transport oversized components. The modular features may be used in conjunction with one or more of the features described above, or in a totally different system.

[0076] An example embodiment of a material-handling robot platform **200** comprises features as described herein is depicted diagrammatically in FIGS. 20A-20D. As shown in these figures, the platform **200** may consist of a robot drive unit **202** with a robot arm **36** and a controller **50**, and one or more base plate assemblies (modules) **204**. As an example, three base plate assemblies **204** are shown in FIGS. 20A-20C. However, more or less than three base plate assemblies **204** could be provided. The controller **50** comprises at least one processor **52** and at least one memory **54** comprising computer program code **56**. The controller **50** is configured to control operations of the various devices and robots of the apparatus **200**. The robot drive unit **202** may comprise features as described above including the linear drive system **30** and the robot drive **34**. However, other types of drives could alternatively be used. Also, any suitable robot arm assembly could be provided.

[0077] The base plate assemblies **204** might be identical, or could have different features. Each base plate assembly **204** may comprise a base plate **206** and various modules **208** that may facilitate traversing motion of the robot drive unit **202** along each base plate assembly **204**. For instance, these modules **208** may include rails (such as, for example, for mechanical linear bearings or magnetic bearings), power delivery arrangements (such as, for example, a service loop, arm, contactless power coupling modules), communication arrangements (such as, for example, a service loop, arm, wireless radio frequency modules, optical communication modules, communication over a contactless power coupling), linear electric motor (such as, for example, a magnet track, stator segments, etc.) and linear position encoder (such as, for example, a scale for an optical, magnetic, inductive or capacitive position encoder read-head). The modules may be contained entirely on individual baseplate assemblies **204**. Alternatively, the modules may be installed so that they span across two or more adjacent baseplates.

[0078] The transport chamber **15** may be provided with a linear encoder **156** as shown in FIG. **10** to sense the linear location of the robot enclosure **28** on the rails **32**. The linear position encoder may comprise one continuous position track, or multiple in-line position tracks and using multiple read heads. All communications with the components in and/or on the robot enclosure **28** may be done optically or wirelessly, or through a power coupling, such that no communication electrical wires or optical wires need to traverse the area **17**. The sensor(s) may comprise a position encoder, such as an optical, magnetic, inductive or capacitive position encoder, a laser interferometer or any other suitable device capable of measuring the position of the robot drive unit along the desired direction of traversing motion, such as relative to the chamber **15** for example. Utilizing the measurements from the position sensor(s), the force produced by the linear actuator may be used to control the position of the robot drive unit along the direction of the desired traversing motion of the robot. In order to support a modular design of the vacuum chamber (i.e., a vacuum chamber composed of multiple sections), an additional encoder read-head for the linear actuation system and/or additional position sensors for the magnetic support system may be utilized to enable smooth transition between individual sections of the vacuum chamber. Specifically, in the case of mechanical linear bearings, the stationary rail may extend by a small extent outside the base plate **206** to allow smooth transfer in the case of rectangular arrangement of linear tracks. Other arrangements, such as triangular, may not need this feature.

[0079] The base plate assemblies **204** may be configured so that they can be arranged next to each other. This arrangement is configured to allow the robot drive unit **202** to traverse from one base plate assembly **204** to another base plate assembly **204** as illustrated by arrow B. To that end, each base plate assembly **204** may feature an alignment arrangement **210**, such as shown in FIG. **20D** for example, to allow two neighboring base plate assemblies **204** to be vertically and laterally aligned at joints **211** and provide smooth transition of the robot drive unit **202** between the two neighboring base plate assemblies **204**. As an example, the alignment arrangement **210** may be formed by precision mating surfaces of a suitable geometry, or by a system of pins and cylindrical and/or oval holes.

[0080] The base plate assembly **204** may further include a sealing feature **212** that may facilitate a sealed joint at the joint **211** when installed into a vacuum chamber. As an example, the sealing feature **212** may include a polished sealing surface, a coated sealing surface, a groove for a sealing element and/or a sealing element, such as an O-ring.

[0081] The base plate assembly **204** may further include a mechanical connection feature **214** between the base plate assembly **204** and a vacuum chamber, such as holes, threaded studs, threaded fasteners or clamps for example. The mechanical connection feature **214** and the base plate **206** itself may be configured so that the various modules **204** that may facilitate traversing motion of the robot drive unit **202** can directly line up with the corresponding modules on the neighboring base plate assembly **204**.

[0082] Referring also to FIG. **20E**, the base plate **206** may be comprises of metal with the rails **32**

integrally formed therewith. Alternatively, the rails may be separately formed and subsequently connected to the top side **207** of the base plate **206**. In this example, the modules **208** are attached to the top side **207** of the base plate **206**. The modules may comprise, for example, a plurality of the magnetic drivers **72** and a plurality of the power coupling **78**; aligned along the longitudinal length of the base plate **206**. As illustrated in FIG. **20F**, the base plate **206** may comprise apertures **209**, **211** therethrough between the top side **207** and the bottom side **213**. Electrical wires **215**, **217** from the modules **72**, **78** may pass through the apertures **209**, **211** to the controller **50** or other electrical supply/controllers. Seals **219**, **221** may be provided at the apertures **209**, **211** to seal apertures **209**, **211**.

[0083] As illustrated in FIGS. **21A-21C**, multiple base plate assemblies **204** may be integrated into a vacuum chamber **220**. In the example of FIGS. **21A-21C**, three base plate assemblies **204** are used. As shown in the figure, the vacuum chamber **220** may include one or more transverse ribs **222** to provide structural strengths. The chamber may also be fitted with one or more lids **224**. As a feature of the present invention, the above described design of the base plate assemblies **204** allows for the base plate assemblies to be installed from the top of the chamber as indicated by arrow C so that each base plate assembly directly faces another base plate assembly without interruptions due to the transverse ribs **222**; thus facilitating uninterrupted traversing motion of the robot drive unit **202** along the entire length of the vacuum chamber **220** as indicated by arrow B.

[0084] In this embodiment the base plate **206** has ledges **223** (see FIGS. **20F** and **20A** for example) at the bottom side at the opposite lateral sides and the opposite ends. The frame of the vacuum chamber has apertures through its bottom side as shown best in FIG. **21A**. The base plates **206** may be placed at the apertures to close the apertures and form at least part of the bottom wall of the vacuum chamber. The ledges **223** may sit on top of bottom portions of the vacuum chamber frame with seals **225** (see FIG. **20F**) forming a seal at the joints.

[0085] In order to further avoid fabrication and transportation of oversized components, the vacuum chamber may be divided into sections **240**, as illustrated in FIGS. **22A-22C**. In the example of FIGS. **22A-22C**, three sections **240** are shown. The chamber sections may be fitted with one or more lids **224**. Each section **240** may utilize one or more base plate assemblies **204** installed in the manner described above with respect to FIGS. **21A-21C**. The sections **240** may include connecting and sealing features **242** at the joints of the sections **240** such as **212** in FIG. **20D** for example. The sealing features may also be used perhaps at other areas, such as the ends of the chamber. This allows one section **240** to be attached to the next section **240** with a seal therebetween around the entire interior open area; a type of square or rectangle in this example, such as seen by the shape in FIG. **22B**. As an example, flanges with fasteners or clamps may be utilized for mechanical connection. Polished sealing surfaces, coated sealing surfaces, grooves for sealing elements and/or various sealing elements, such as O-rings, may be used to achieve proper sealing. These are merely some examples and should not be considered limiting.

[0086] It is noted that as the X axis motion range for the linear vacuum robot becomes very large, it becomes challenging to build a position track for the position sensor which is a part of the linear actuator system described earlier. To overcome this limitation, a segmented position track approach with dual position sensors is disclosed here. With reference to FIG. **24**, as the robot drive traverses in the X direction, both positions sensors RH1, RH2 generate valid position data offset by spacing length F at all instances except when one or the other of the sensors RH1, RH2 is targeting the transition between the ends of the two position tracks E1, E2. Note the gap or transition N. The disclosed solution overcomes the period at which one sensor data is invalid by switching over to the other sensor. Since the sensors RH1, RH2 are physically offset relative to the length of the guides/tracks E1, E2, at least one of the sensors RH1, RH2 always reports valid position. Since the location of the transition N between tracks E1, E2 is a design quantity, it is known ahead of time and the controller **2406** makes a decision based on the known location of the transition N to switch between the sensors RH1, RH2. In other embodiments, algorithms may be implemented to use,

merge or combine the data from the sensors RH1, RH2 to achieve a gradual or smooth transition of data as the active sensor is switched. During periods of time where data from both sensors is valid, the data from both may be combined or merged in various manners to achieve higher accuracy, resolution, reliability, or noise rejection.

[0087] It is noted that as the X axis motion range for the maglev supported robot becomes very large, it becomes challenging to build smooth and continuous surfaces for the gap sensors to target. With reference to FIG. 31, there are two vertical sensors P, Q targeted at target T1, T2. There is a break/transition in the targets T1, T2 at location L. The gap measurement from either sensor is valid as long as they are not traveling through the location L. The controller 2406 makes a decision based on X motion location from sensor RH1, RH2 on time instants when the gap sensor is transitioning through L. The informed algorithm is implemented on the controller 2406 to use the gap data from P or Q.

[0088] An example embodiment may be provided with an apparatus comprising: a first base plate, where the first base plate is configured to have at least one linear drive component and/or at least one power coupling component connected to a top side of the first base plate, where the first base plate is configured to be located inside a vacuum chamber; and a plurality of rails or transport guides on the top side of the first base plate; where an end of the first base plate comprises at least one alignment feature configured to align an end of the first base plate to an end of a second base plate, where the first base plate is configured to provide, in combination with the second base plate, a structural platform inside the vacuum chamber for a robot drive to move in the vacuum chamber along the plurality of transport guides.

[0089] The at least one alignment feature may be configured to align the plurality of transport guides with transport guides of the second base plate. The at least one alignment feature may comprise at least one projection extending from the end of the first base plate and at least one recess extending into the first end of the first base plate. The end of the first base plate may comprise at least one mechanical connection feature projecting from the end of the first base plate, where the at least one mechanical connection feature is configured to be mechanically attached to the second base plate with at least one fastener. The first base plate may be configured to be mechanically attached directly to the vacuum chamber at a bottom side of the vacuum chamber. The first base plate may be configured to form at least a portion of a bottom wall of the vacuum chamber. The first base plate may close an aperture through the bottom of the vacuum chamber. The first base plate may comprise first ledges on two opposite lateral sides of the first base plate, where the first ledges are configured to sit on top of shelves formed by the vacuum chamber on opposite lateral side of the vacuum chamber. The first base plate may comprise second ledges on two opposite ends of the first base plate, where the second ledges are configured to sit on top of shelves formed by the vacuum chamber extending transversely across the vacuum chamber. The first base plate may comprise apertures therethrough configured to have electrical wires extend through the first base plate from a bottom side of the first base plate of the top side of the first base plate, and the apparatus further comprises seals configured to seal the apertures with the wires. The apparatus may further comprise the vacuum chamber, where the vacuum chamber comprises a plurality of main sections configured to be attached to one another in an end-to-end configuration to form an elongate shaped frame having the first base plate connected thereto, where the main sections comprise opposite lateral side walls and a bottom wall portion, and at least one lid section configured to be removably connected to top sides of the opposite lateral side walls. The apparatus may further comprise the at least one linear drive component connected to the top side of the first base plate, where the at least one linear drive component is configured to provide a magnetic field to move the robot drive along the plurality of transport guides. The apparatus may further comprise the power coupling component, where the power coupling component is configured to transfer power to another power coupling component on the robot drive, where the transfer of power is with induction.

[0090] An example method may be provided comprising: connecting a first base plate assembly inside a vacuum chamber, where the first base plate assembly comprises a first base plate and a first plurality of rails or transport guides on a top side of the first base plate; and connecting a second base plate assembly inside the vacuum chamber, where the second base plate assembly comprises a second base plate, where an end of the second base plate is connected to an end of the first base plate, where the first base plate assembly comprises at least one alignment feature configured to align the end of the first base plate to the end of the second base plate, where the first base plate is configured to provide, in combination with the second base plate, a structural platform inside the vacuum chamber for a robot drive to move in the vacuum chamber along the first plurality of transport guides.

[0091] The second base plate assembly may comprise a second plurality of rails or transport guides on a top side of the second base plate, where the connecting of the second base plate assembly inside the vacuum chamber comprises the at least one alignment feature causing aligning of the second plurality of transport guides with the first plurality of transport guides. The connecting of the first base plate assembly inside a vacuum chamber may comprise the first base plate forming at least a portion of a bottom wall of the vacuum chamber. The first base plate may close an aperture through the bottom of the vacuum chamber. The connecting of the first base plate assembly inside the vacuum chamber may comprise the first base plate including first ledges on two opposite lateral sides of the first base plate, where the first ledges sit on top of shelves formed by the vacuum chamber on opposite lateral side of the vacuum chamber. The connecting of the first base plate assembly inside the vacuum chamber may comprise the first base plate including second ledges on two opposite ends of the first base plate, where the second ledges sit on top of shelves formed by the vacuum chamber extending transversely across the vacuum chamber.

[0092] An example method may be provided comprising: providing a base plate; providing a plurality of rails or transport guides on a top side of the base plate; providing an end of the base plate with at least one alignment feature configured to align the end of the first base plate to an end of a second base plate; connecting to the top side of the base plate at least one of: a power coupling component configured to transfer power to another power coupling component on a robot drive, where the transfer of power is with induction, or a linear motor component configured to provide a magnetic field to move the robot drive along the plurality of transport guides, where the base plate is configured to be connected inside a vacuum chamber with the second base plate to provide a structural platform inside the vacuum chamber for the robot drive to move in the vacuum chamber along the first plurality of transport guides.

[0093] An example embodiment may be provided in an apparatus comprising means for connecting a first base plate assembly inside a vacuum chamber, where the first base plate assembly comprises a first base plate and a first plurality of rails or transport guides on a top side of the first base plate; and means for connecting a second base plate assembly inside the vacuum chamber, where the second base plate assembly comprises a second base plate, where an end of the second base plate is connected to an end of the first base plate, where the first base plate assembly comprises at least one alignment feature configured to align the end of the first base plate to the end of the second base plate, where the first base plate is configured to provide, in combination with the second base plate, a structural platform inside the vacuum chamber for a robot drive to move in the vacuum chamber along the first plurality of transport guides.

[0094] An alternative means to reduce fabrication and transportation of oversized components is to eliminate the baseplate **204** of FIGS. **20-21**. FIGS. **23A-23C** show an example of an embodiment where the base plate is an integral part of the chamber **241**. The chamber may be one-piece or segmented pieces as illustrated in FIG. **23A**. All the features and components required to facilitate traversing motion of the robot drive unit **202** may be installed directly on the floor (or walls) of the chamber **241**, such as rails or transport guides (such as, for example, for mechanical linear bearings or magnetic bearings), power delivery arrangements (such as, for example, a service loop, arm,

contactless power coupling modules), communication modules (such as, for example, a service loop, arm, wireless radio frequency modules, optical communication modules, communication over a contactless power coupling), linear electric motor (such as, for example, a magnet track, stator segments, etc.) and linear position encoder (such as, for example, a scale for an optical, magnetic, inductive or capacitive position encoder read-head). The chamber sections **241** may include connecting and sealing features **242** at the joints of the sections **241**. The sealing features may also be used perhaps at other areas, such as the ends of the chamber. This allows one section **241** to be attached to the next section **241** with a seal therebetween around the entire interior open area; a type of square or rectangle in this example, such as seen by the shape in FIG. **23B**. As an example, flanges with fasteners or clamps may be utilized for mechanical connection. Polished sealing surfaces, coated sealing surfaces, grooves for sealing elements and/or various sealing elements, such as O-rings, may be used to achieve proper sealing. These are merely some examples and should not be considered limiting.

Linear Vacuum Robot Control System Architecture

[0095] A block diagram of an example embodiment of the linear robot control system is provided in FIG. **25**.

[0096] The master controller may perform the following functions: the user interface, communication with host controller (e.g., utilizing serial communication or Ethernet communication), configuration data management, high-level motion planning (i.e., sequencing of moves of the robot), trajectory generation (calculation of motion profiles for each move for each axis of motion), position control for all axes of motion and APS (adaptive placement system). One example of an adaptive placement system is described in U.S. Pat. No. 10,058,996 which is hereby incorporated by reference in its entirety.

[0097] The master controller may receive from the host controller various commands, including configuration, request and action commands, e.g., commands to perform a pick or place operation, and it may report back to the host controller completion of the commands and other information.

[0098] The master controller may receive over the high-speed network positions of all axes of motion (periodically from the motor amplifiers), status of digital and analog inputs (from the I/O module and, if applicable, from the motor amplifiers) and timing of digital input changes (from the I/O module and, if applicable, from the motor amplifiers). The master controller may send over the high-speed network commanded currents for each axis of motion (periodically to the motor amplifiers) and information to set digital and analog outputs (to the I/O module and, if applicable, to the motor amplifiers). Whether a rail based linear system or a maglev linear systems, the control system may comprise use of an optical communications link or other wireless communications link. For example, as shown in FIG. **25** the system may comprise two optical communications modules with respective viewports in the vacuum chamber wall and the robot drive wall. Thus, an optical communication link may be provided between the external controller and the robot drive, which may include a closed loop control as noted above.

[0099] The I/O module may read digital and analog inputs (which may include inputs from APS sensors) and set digital and analog outputs. The I/O module may receive over the high-speed network information to set digital and analog outputs (from the master controller), and it may send over the high-speed network status of digital and analog inputs (to the master controller) and timing of digital input changes (also to the master controller).

[0100] Each of the motor amplifiers may perform the following functions: execution of motor commutation algorithm(s), execution of current control loops, reading digital and analog inputs, and setting digital and analog outputs. Each of the motor amplifiers may read periodically measured position(s) from the position encoder(s) and set output voltages for control of the motor(s). Each motor amplifier may receive from the master controller over the high-speed network commanded current(s) for the axis or axes of motion supported (periodically) and information to set digital and analog outputs. It may send to the master controller over the high-

speed network the measured positions of the axis or axes of motion supported (periodically), status of digital and analog inputs and, if applicable, timing of digital input changes.

[0101] The high-speed network (e.g., EtherCAT) may facilitate communication between the master controller and the I/O module as well as the motor amplifiers. The outbound traffic (i.e., from the master controller to the I/O module and motor amplifiers) may include commanded currents for each axis of motion (sent periodically from the master controller to the motor amplifiers) and information to set digital and analog outputs (sent from the master controller to the I/O module and, if applicable, to the motor amplifiers). The inbound traffic (i.e., from the I/O module and motor amplifiers to the master controller) may include measured positions (from the motor amplifiers), status of digital and analog inputs (from the I/O module and, if applicable, from the motor amplifiers) and timing of digital input changes (from the I/O module and, if applicable, from the motor amplifiers).

[0102] If APS (adaptive placement system) functionality is required, APS sensor(s) may be routed directly or through an I/O connection board to one or more inputs of the I/O module. The purpose of the optional I/O connection board is to reduce the number of inputs routed to the I/O module.

[0103] A block diagram of another example embodiment of the linear robot control system is provided in FIG. 26. In this example embodiment, the master controller may be located in the robot drive as opposed to being part of the external controller.

[0104] In the example embodiment of FIG. 26, the master controller may communicate with the motor amplifiers over the high-speed network substantially the same way as in the previous embodiment. However, another means of communication may be used for communication between the master controller and the I/O module. As an example, a separate communication network (e.g., Ethernet) may be used. The same communication network may also be utilized for communication with the host controller, as depicted diagrammatically in FIG. 26, in which case a network router may be conveniently incorporated into the external controller board. Alternatively, a separate means of communication may be used for communication between the host controller and the master controller and for communication between the master controller and the I/O module. These two communication channels may take place over the same physical medium (e.g., serial communication over Ethernet) or use different physical media.

[0105] The communication between the master controller and the I/O module may allow for synchronization of the clocks running on the two devices or feature another mechanism to properly determine timing of digital input changes on the I/O module for APS calculation purposes (for instance, an offset between the two clocks may be identified periodically and applied when a digital input change occurs).

[0106] In yet another example embodiment, the high-speed communication via bidirectional light beam may be routed through the power coupling. The power coupling may employ the same set of coils as for power transmission or and additional set of coils to transmit data. An example embodiment equivalent to FIG. 25 is depicted diagrammatically in FIG. 27, and an example embodiment equivalent to FIG. 26 is shown diagrammatically in FIG. 28.

[0107] Block diagrams of additional example embodiments of the linear robot control system that include control of a magnetic support system (maglev) are shown in FIGS. 29-30.

[0108] The maglev controller may perform position control of the robot drive (e.g., control the five degrees of freedom associated with the lateral position, vertical position, pitch angle, roll angle and yaw angle of the robot drive) and run current control loops for each of the actuators of the magnetic support system. In the process, the maglev controller may periodically read measured positions from the position sensors of the magnetic support system (e.g., two horizontal sensors and three vertical sensors) and set output voltages for the force actuators of the magnetic support system (e.g., two horizontal actuator pairs and three vertical actuator pairs). The maglev controller may receive from the master controller over the high-speed network various commands, including commands to lift off, to maintain a given position (which may be conveniently expressed in the

form of gaps between the robot drive and the guides of the magnetic support system) and to land the robot drive. Alternatively, the maglev controller may receive from the master controller a stream of commanded positions (e.g., in the form of data frames sent periodically).

[0109] As another alternative, the master control may execute position control of the robot drive via the magnetic support system. In the process, the master controller may receive periodically from the maglev controller over the high-speed network measured positions from the sensors of the magnetic support system and send periodically to the maglev controller over the high-speed network commanded currents for each force actuator of the magnetic support system. In this arrangement, the maglev controller may still run current control loops for each of the actuators of the magnetic support system.

[0110] In order to support a modular design of the vacuum chamber (i.e., a vacuum chamber composed of multiple sections), an additional encoder read-head for the linear actuation system and/or additional position sensors for the magnetic support system may be utilized to enable smooth transition between individual sections of the vacuum chamber. This is diagrammatically illustrated in FIGS. 29 and 30.

[0111] All of the above embodiments may include additional feature that have been left from the diagrams for the sake of simplicity. For example, the external controller may include support for a teach pendant, e-stop, interlocks, safety circuitry (including solid-state components and electromechanical contactor(s)) and energy storage (e.g., batteries and/or capacitors). Similarly, the on-board controller may feature safety circuitry (including solid-state components and electromechanical contactor(s)) back emf regeneration system and power storage (e.g., batteries and/or capacitors).

[0112] Although the above example embodiments described with respect to FIGS. 25-28 show three rotary axes of motion (T1, T2 and T3), any number of rotary axes or no rotary axis may be used. Similarly, although the above example embodiments of FIGS. 25-28 show one z-axis, any number of z-axes or no z-axis may be employed.

[0113] It should be understood that the foregoing description is only illustrative. Various alternatives and modifications can be devised by those skilled in the art. For example, features recited in the various dependent claims could be combined with each other in any suitable combination(s). In addition, features from different embodiments described above could be selectively combined into a new embodiment. Accordingly, the description is intended to embrace all such alternatives, modifications and variances which fall within the scope of the appended claims.

Claims

1. A transport apparatus, comprising: a first base plate having a first position track, where the first base plate is configured to have at least one first linear drive component and/or at least one first power coupling component connected to a top side of the first base plate, where the first base plate is configured to be located inside a vacuum chamber; a second base plate having a second position track, where the second base plate is configured to have at least one second linear drive component and/or at least one second power coupling component connected to a top side of the second base plate, where the second base plate is configured to be located outside the vacuum chamber, where an end of the second base plate is aligned with an end of the first base plate and an end of the second position track is aligned with an end of the first position track to form a gap between the second position track and the first position track; a plurality of transport guides on the top side of the first base plate and on the top side of the second base plate; and a robot drive configured to move on the plurality of transport guides along the first position track and the second position track, the robot drive comprising a first position sensor configured to sense the first position track and the second position track and a second position sensor configured to sense the first position

track and the second position track, the first position sensor being spaced from the second position sensor such that when the first position sensor engages the gap, the second position sensor senses a position of the robot drive on one of the first position track or the second position track and when the second position sensor engages the gap, the first position sensor senses the position of the robot drive on one of the first position track or the second position track, where a position of the robot drive on the first position track or the second position track is used to determine an operation of the robot drive; where an end of the first base plate comprises at least one alignment feature configured to align the end of the first base plate to an end of the second base plate, and where the first base plate comprises a seal between the inside of the vacuum chamber and the outside of the vacuum chamber.

2. The transport apparatus as in claim 1, where the first position sensor is spaced from the second position sensor relative to a length of the first position track or a length of the second position track.

3. The transport apparatus as in claim 1, further comprising a controller on the robot drive, the controller being configured to switch between activation of the first position sensor and the second position sensor based on a location of the gap.

4. The transport apparatus as in claim 3, where an algorithm is implemented to use data from the first position sensor and the second position sensor to obtain a gradual transition of data during the switch between the activation of the first position sensor and the second position sensor.

5. The transport apparatus as in claim 3, where an algorithm is implemented to combine data from the first position sensor and the second position sensor when both the first position sensor and the second position sensor are activated.

6. The transport apparatus as in claim 1, where the at least one alignment feature is configured to align the plurality of transport guides on the first base plate with the plurality of transport guides on the second base plate.

7. The transport apparatus as in claim 1, where the at least one alignment feature comprises at least one projection extending from the end of the first base plate and at least one recess extending into the first end of the first base plate.

8. The transport apparatus as in claim 1, where the end of the first base plate comprises at least one mechanical connection feature projecting from the end of the first base plate, where the at least one mechanical connection feature is configured to be mechanically attached to the second base plate with at least one fastener.

9. The transport apparatus as in claim 1, where the first base plate is configured to be mechanically attached directly to the vacuum chamber at a bottom side of the vacuum chamber.

10. The transport apparatus as in claim 1, further comprising the vacuum chamber, where the vacuum chamber comprises a plurality of main sections configured to be attached to one another in an end-to-end configuration to form an elongate shaped frame having the first base plate connected thereto, where the main sections comprise opposite lateral side walls and a bottom wall portion, and at least one lid section configured to be removably connected to top sides of the opposite lateral side walls.

11. The transport apparatus as in claim 1, further comprising the at least one linear drive component connected to the top side of the first base plate, where the at least one linear drive component is configured to provide a magnetic field to move the robot drive along the plurality of transport guides.

12. The transport apparatus as in claim 1, further comprising the power coupling component, where the power coupling component is configured to transfer power to another power coupling component on the robot drive, where the transfer of power is with induction.

13. A method for producing a transport apparatus, comprising: connecting a first base plate assembly inside a vacuum chamber, where the first base plate assembly comprises a first base plate having a first position track and a first plurality of transport guides on a top side of the first base

plate; and connecting a second base plate assembly inside the vacuum chamber, where the second base plate assembly comprises a second base plate having a second position track and a second plurality of transport guides on a top side of the second base plate, where an end of the second base plate is connected to an end of the first base plate and an end of the second position track is aligned with an end of the first position track to form a gap between the second position track and the first position track; and providing a robot drive configured to move on the transport guides along the first position track and the second position track, the robot drive comprising a first position sensor configured to sense the first position track and the second position track and a second position sensor configured to sense the first position track and the second position track, the first position sensor being spaced from the second position sensor such that when the first position sensor engages the gap, the second position sensor senses a position of the robot drive on one of the first position track or the second position track and when the second position sensor engages the gap, the first position sensor senses the position of the robot drive on one of the first position track or the second position track, where a position of the robot drive on the first position track or the second position track is used to determine an operation of the robot drive; where the first base plate assembly comprises at least one alignment feature configured to align the end of the first base plate to the end of the second base plate, where the first base plate comprises a seal between an inside of the vacuum chamber and an outside of the vacuum chamber, where the first base plate is configured to provide, in combination with the second base plate, a structural platform inside the vacuum chamber for the robot drive to move in the vacuum chamber along the first plurality of transport guides and outside the vacuum chamber for the robot drive to move outside of the vacuum chamber along the second plurality of transport guides.

14. The method of claim 13, further comprising spacing the first position sensor from the second position sensor relative to a length of the first position track or a length of the second position track.

15. The method of claim 13, further comprising providing a controller on the robot drive and using the controller to switch between activation of the first position sensor and the second position sensor based on a location of the gap.

16. The method of claim 15, further comprising implementing an algorithm to use data from the first position sensor and the second position sensor to obtain a gradual transition of data during the switch between the activation of the first position sensor and the second position sensor.

17. The method of claim 15, further comprising implementing an algorithm to combine data from the first position sensor and the second position sensor when both the first position sensor and the second position sensor are activated.

18. A method for producing a transport apparatus, comprising: providing a first base plate having a first position track; providing a second base plate having a second position track, where an end of the second base plate is aligned with an end of the first base plate and an end of the second position track is aligned with an end of the first position track to form a gap between the first position track and the second position track; providing a first plurality of transport guides on a top side of the first base plate and a second plurality of transport guides on a top side of the second base plate; providing the end of the first base plate with at least one alignment feature configured to align the end of the first base plate to the end of the second base plate; providing a robot drive configured to move on the first plurality of transport guides along the first position track and on the second plurality of transport guides along the second position track, the robot drive comprising a first position sensor configured to sense the first position track and the second position track and a second position sensor configured to sense the first position track and the second position track, the first position sensor being spaced from the second position sensor such that when the first position sensor engages the gap, the second position sensor senses a position of the robot drive on one of the first position track or the second position track and when the second position sensor engages the gap, the first position sensor senses the position of the robot drive on one of the first position track

or the second position track, where a position of the robot drive on the first position track or the second position track determines an operation of the robot drive; and connecting to the top side of the first base plate at least one of: a power coupling component configured to transfer power to another power coupling component on the robot drive, where the transfer of power is with induction, or a linear motor component configured to provide a magnetic field to move the robot drive along the plurality of transport guides; where the first base plate comprises a seal between an inside of a vacuum chamber and an outside of the vacuum chamber, where the first base plate is configured to be connected inside a vacuum chamber with the second base plate to provide a structural platform inside the vacuum chamber for the robot drive to move in the vacuum chamber along the first plurality of transport guides and outside the vacuum chamber for the robot drive to move outside the vacuum chamber along the second plurality of transport guides.
